

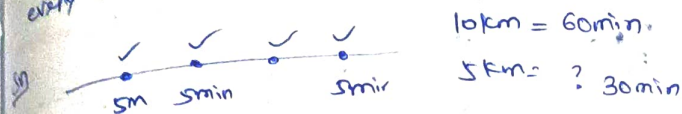
Time and distance

Typ-1

$$\text{Speed (S)} = \frac{\text{Distance (D)}}{\text{time (T)}}$$



man running $\rightarrow 10 \frac{\text{km}}{\text{h}}$, took rest 5min every 1km and travelled 5km

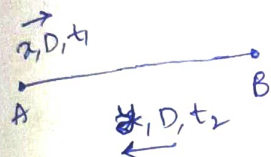


Total time = 50 min $\leftarrow$ travel time $\leftarrow 30 + 20 \rightarrow$ brake time

Typ-2

$$\text{Avg Speed} = \frac{\text{Total dist}}{\text{Total time}}$$

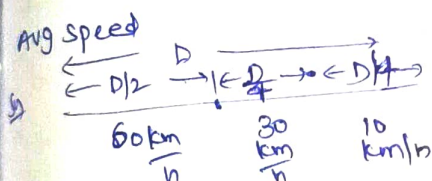
(or) $\frac{\text{Sum of Individual dist}}{\text{Sum of Individual time}}$



man travelled
A $\rightarrow$ B
B
A $\leftarrow$ B

$$\text{Avg Speed} = \frac{T \cdot D}{T \cdot t} = \frac{D + D}{t_1 + t_2} = \frac{2D}{\frac{D}{x} + \frac{D}{y}} = \frac{2xy}{x+y}$$

Ex: A man travelled $\frac{1}{2}$ Journey train @ $60 \frac{\text{km}}{\text{h}}$, $\frac{1}{2}$ of remaining bus @ $30 \frac{\text{km}}{\text{h}}$, rest by cycle @ $10 \frac{\text{km}}{\text{h}}$



$$\text{Avg Speed} = \frac{D}{\frac{D}{2} + \frac{D/4}{30} + \frac{D/4}{10}} = \frac{D}{\frac{D}{120} + \frac{D}{120} + \frac{D}{40}} = \frac{D}{\frac{1+1+3}{120}} = 24 \frac{\text{km}}{\text{h}}$$

Ex: Car $\frac{1}{2}$ Journey at $36 \frac{\text{kmph}}$, rest at $24 \frac{\text{kmph}}$ avg Speed = $\frac{2xy}{x+y} = \frac{2 \times 36 \times 24}{(36+24)} = 28.8$

Type 5: Ex: A boy going School from home @ $3 \frac{\text{kmph}}$ reach 10 late, next day increased Speed by $1 \frac{\text{kmph}}$ reach School 20min early. Find dist b/w two places?

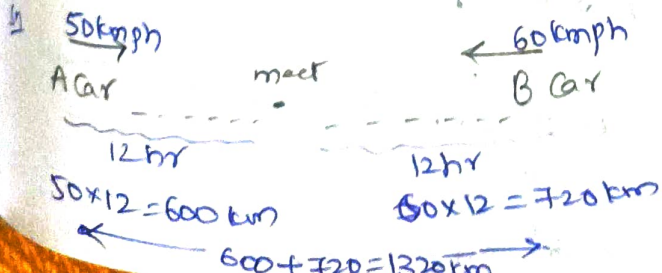
$$D = 3 \left\{ \frac{t+10}{60} \right\}, D = 4 \left\{ \frac{t-20}{60} \right\}, \frac{3(t+10)}{60} = \frac{4(t-20)}{60}$$

$S_1 = 3 \frac{\text{kmph}}, t+10$ School
 $S_2 = 4 \frac{\text{kmph}}, t-20$
Dist = S.T

$$D = S \cdot t \Rightarrow \frac{3(10+10)}{60} [t=110] \Leftarrow 3t+30 = 4t-80$$

$D = 6 \text{ km}$

Type 6: Two Car A & B travel towards each other at $50 \frac{\text{kmph}}$ & $60 \frac{\text{kmph}}$, at time they meet second car travelled 120km more than first, dist b/w A & B is -



$60 - 50 = 10 \frac{\text{kmph}}$ so
For 1hr = 10 km
? $\times 120 \text{ km}$

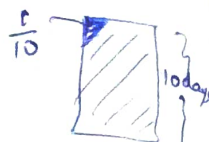
$$\frac{120}{10} = 12 \text{ hr}$$

Time and Work [20 Times]

Relation b/w Time & work

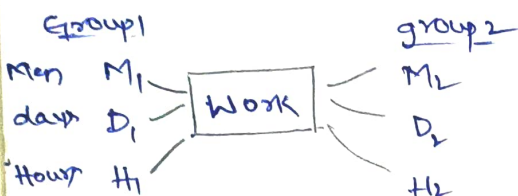
$$T \propto \frac{1}{W}$$

A can do a piece of work in 10 days then A's 1 day work is $\frac{1}{10}$



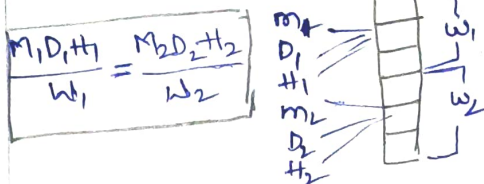
Type 1: chain rule

If work is same



$$M_1 D_1 H_1 = M_2 D_2 H_2$$

Different works



$$\frac{M_1 D_1 H_1}{W_1} = \frac{M_2 D_2 H_2}{W_2}$$

Efficiency also given

$$E_1 \rightarrow M_1 D_1 H_1 \times \% \rightarrow W_1$$

$$E_2 \rightarrow M_2 D_2 H_2 \times \% \rightarrow W_2$$

$$\frac{M_1 D_1 H_1 \times \%}{W_1} = \frac{M_2 D_2 H_2 \times \%}{W_2}$$

Ex: 60 men construct wall @ 250 days
they worked all but after 200 days
work stopped for 10 days then how
many extra men req. to complete on time

Sol: $60(250) = 60(200) + 40(60+x)$
 $60(250-200) = 40(60+x)$
 $x = 15 \text{ days}$

Ex: A contract completed in 56 days
104 men @ 8 hr/day after 30 days
 $\frac{2}{5}$ of work done. How many extra
men employed to complete work on
time now working 9 hr/day!

Sol: $\frac{104 \times 30 \times 8}{\frac{2}{5}} = \frac{(104+x) \times 26 \times 9}{\frac{3}{5}}$
 $x = 56$

Ex: 7 machines take 7 min to make 7 toys same
rate 100 machines to make 100 toys

Sol: $\frac{7 \times 7}{7 (\text{toys})} = \frac{100 \times 2}{100 (\text{toys})}$
 $x = 2$

Type 1, 3, 4, 5 A $\rightarrow x$ days

B $\rightarrow y$ day
A & B together = $\left(\frac{1}{x} + \frac{1}{y}\right)$ 1 day together A & B work

A $\rightarrow x$

A & B $\rightarrow y$

B $\rightarrow ? = \frac{1}{y} - \frac{1}{x}$

A $\rightarrow x$

B $\rightarrow y$

C $\rightarrow z$

A, B, C = $\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$ 1 day together A, B, C work

A & B $\rightarrow x$

B & C $\rightarrow y$

A & C $\rightarrow z$

A, B, C = ?

$2(A+B+C) = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$

$A+B+C = \frac{1}{2} \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}\right)$

Ex: A can work complete = 36 days
B can complete in 12 day then
A & B together can do —

Sol: A $\rightarrow 36$ days
B $\rightarrow 12$ days

$A+B = \frac{1}{36} + \frac{1}{12} = \frac{1+3}{36} = \frac{4}{36}$
 $= \frac{1}{9}$ ✓ A & B 9 days

Ex: A can do work 16 days more to complete work by A & B
B take 4 days more " " by A & B then
A & B together can complete in —

Sol: A & B = x days, A = $x+16$, B = $x+4$

$\left(\frac{1}{x}\right) = \frac{1}{(x+16)} + \frac{1}{(x+4)} \Rightarrow \frac{1}{x} = \frac{(x+4) + (x+16)}{(x+16)(x+4)}$
 $A+B = \frac{1}{x+16} + \frac{1}{x+4}$

Type 7: wages Concept

Ex: A, B, C together did work earned 575

A, B, C supposed to finish $\frac{19}{23}$ yr work together
the B share

Sol $\Rightarrow \frac{4}{23} \times 575$

Type 8: A, B, C $\rightarrow x$

B, C $\rightarrow y$

A, B, C $\rightarrow z$

$$2A = \left(\frac{1}{x} + \frac{1}{z}\right) - \frac{1}{y}$$

$$A = \frac{1}{2} \left\{ \frac{1}{x} + \frac{1}{z} - \frac{1}{y} \right\}$$

Type 10:

A works twice B then B = 2x days
A = x days

A works 2x. more efficient than B

work ratio $100+x : 100$

days ratio $\frac{1}{100+x} : \frac{1}{100}$

Ex: A is twice good as B together they can do in 18 days. If A alone has many days

Sol A = x days, B = 2x days

$$\frac{1}{18} = \frac{1}{x} + \frac{1}{2x} \Rightarrow \frac{2x+x}{2x^2} = \frac{3x}{2x^2} \Rightarrow 2x = 3x + 18 \Rightarrow \boxed{x = 27 \text{ days}}$$

Type 11 Ex: 1200m, 500w can build bridge in 2 weeks, 900m & 250w took 3 weeks then
how many men req to complete in 1 week

Sol: $1200m + 500w = \frac{1}{2} \times 1$

$$900m + 250w = \frac{1}{3} \times 2$$

$$1200m + 500w = \frac{1}{2}$$

$$1800m + 500w = \frac{2}{3}$$

$$600m = \left(\frac{2}{3} - \frac{1}{2}\right) \Rightarrow 600m = \frac{1}{6} \text{ weeks}$$

then 600m = 6 weeks

x = 1 week

$$600 \times 1 = 6x$$

$$\boxed{x = 3600 \text{ men}}$$

Type 9: 5 Skilled Can do in 20 days, 8 semi skilled - 25, 10 unskilled = 30

then 2A, 6B, 5C group Can do in

Sol $5A = 20$ $8B = 25$ $10C = 30$ $2A + 6B + 5C \leftarrow \text{Team}$
 $A = \frac{20}{5}$ $B = \frac{25}{8}$ $C = \frac{30}{10}$ $\Rightarrow 2\left(\frac{20}{5}\right) + 6\left(\frac{25}{8}\right) + 5\left(\frac{30}{10}\right)$

$5Sk = 20$ $8sem = 25$ $10Un = 30$ $\Rightarrow \frac{8}{1} + \frac{3 \times 25}{4} + \frac{30}{2}$
 $1Sk = 100$ $1sem = 200$ $1Un = 300$ $= \frac{(4 \times 8) + (75) + 60}{4}$
 $\frac{1 \text{ day work}}{100}$ $\frac{1 \text{ day work}}{200}$ $\frac{1 \text{ day work}}{300}$ $= \frac{32 + 75 + 60}{4}$

$$2Sk + 6sem + 5Un = 2\left(\frac{1}{100}\right) + 6\left(\frac{1}{200}\right) + 5\left(\frac{1}{300}\right)$$

$$= \frac{12 + 18 + 10}{600} = \frac{40}{600} = \frac{1}{15} \text{ days}$$

Typ: 12: Based on remaining work

1. A = 20 days work starts with A done 5 days work only, remaining done by B
B = 30 days then, find no. of days to complete work

$$\text{Q1 } 5 \left\{ \frac{1}{20} \right\} + x \left\{ \frac{1}{30} \right\} = 1 \Rightarrow \frac{15+2x}{60} = 1 \Rightarrow x = \frac{45}{2} \Rightarrow 22\frac{1}{2} \text{ days}$$

$\downarrow$
 A's 1 day work B's 1 day work B's total work days

Total = $5 + 22\frac{1}{2} = 27\frac{1}{2}$

Q2) A machine P $\frac{1 \text{ lakh book print in } 8 \text{ hrs}}$
 " Q $\frac{12}{10 \text{ hrs}}$
 " R $\frac{12}{12 \text{ hrs}}$

Individual

while all mk start at 9 am while p closed at 11 am and remaining 2mk completed work approximate finished time

$$\text{Q3 } 2 \left\{ \frac{1}{8} + \frac{1}{10} + \frac{1}{12} \right\} + x \left\{ \frac{1}{10} + \frac{1}{12} \right\} = 1 \Rightarrow \frac{2(15+12+10) + x(12+10)}{120} = 1 \Rightarrow 2x = 120 - 70$$

$\Rightarrow 9 \text{ am} + 2 \text{ hr} + 2 \text{ hr} \Rightarrow \text{final } 1 \text{ pm finished}$
 $\downarrow \quad \quad \downarrow$
 all mkr Q, R

- 3) A, B, C individually can do work in 10, 12, 15 days respectively. They started work together but A left the work 5 days before the completion, B also left 2 days after A left then total work days = ?

$$\text{Q4 } x(A+B+C) + 2(B+C) + 3(C) = 1$$

$$x \left\{ \frac{1}{10} + \frac{1}{12} + \frac{1}{15} \right\} + 2 \left\{ \frac{1}{12} + \frac{1}{15} \right\} + 3 \left\{ \frac{1}{15} \right\} = 1$$

$$\frac{x[6+5+4] + 2[5+4] + 4(3)}{60} = 1$$

$$15x + 18 + 12 = 60$$

$$x = 2$$

$$x + 2 + 3 = \text{Total}$$

$$2 + 2 + 3 = \text{Total}$$

$$\boxed{\text{Total} = 7 \text{ days}}$$

Profit and loss

Typ 1: $\text{profit \%} = \frac{P}{CP} \times 100$

$\text{loss \%} = \frac{L}{CP} \times 100$

✓ Type 2:

~~Item~~ Sold @ 810 and got 10% loss on sp. what S.P. he would gain 20% profit. $\Rightarrow 90\% = 810$

$120\% = ?$ (880)

Typ 3: Cost price (Cp) of 10 Apples is equal to selling price of 8 Apples. Find profit (or) loss %

Sol Cp of 10 Apples = 100/-

1 Apple = $\frac{100}{10} = 10/-$

Sp of 8 Apples = 100/-

1 Apple = $\frac{100}{8} = 12.5\%$

profit % = $\frac{2.5}{10} \times 100 = 25\%$

Short cut

$CP(10) = SP(8)$

$\frac{CP}{SP} = \frac{8}{10}$ diff 2 $\frac{2}{8} \times 100 = 25\%$

Typ 5: Cp of 11 Apples is 10/- and Sp of 10 Apples is 11/- Find profit (or) loss %

Sol Cp of 11 Ap = 10/-

1 Ap = $\frac{10}{11}$ /-

Sp of 10 Ap = 11 /-

1 Ap = $\frac{11}{10}$ /-

profit $\frac{\frac{11}{10} - \frac{10}{11}}{\frac{10}{11}} \times 100 = 21\%$ (↑)

Typ 6: A Shop Keeper Sold an item @ 10% loss if he had sold for excess amount of 20/- then he got 20% profit on C.P. Find C.P.

Sol 90% $\rightarrow$ S.P. $x/-$

120% $\rightarrow$ $x + 20/-$

30% $\rightarrow$ 20%.

100% $\rightarrow$? 400

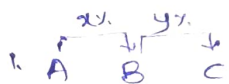
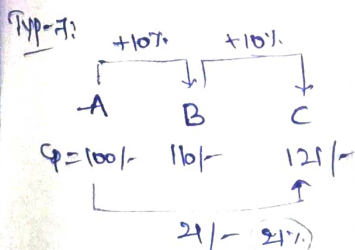
Short cut (Sol) 30% = 20/-

100% = ? 400

Ex: Anith sold profit @ 11%, Had she sold it ₹ 175 more she gained 18% then cp of painting is -

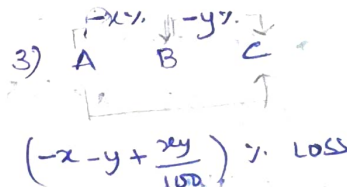
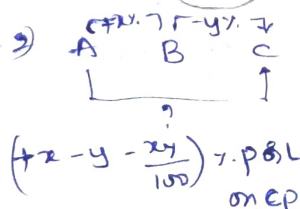
Sol 7% = 175

100% = ? 2500



$(x + y + \frac{xy}{100})\%$ (↑) P on CP

$10 + 10 + \frac{10 \times 10}{100} = 21\%$



* got 10% m B, B got 10%

profit on B then finally

* got % m C

Typ 8: Remaining on (Rest on) successive disc (s) P

If after successive discount of 30%, 20% and 10% a customer pay ₹ 252

What is list price of article Sol: $252 = CP \times \frac{100-30}{100} \times \frac{100-20}{100} \times \frac{100-10}{100} \Rightarrow CP = 500$

S.P = $CP \times \frac{100 \pm x}{100} \times \frac{100 \pm y}{100} \times \dots$

Typ 10: Dishonest dealer Concept

1. A dealer used 1200 gm instead of 1150 gm. Find profit or loss?
2. dealer used 900 gm instead of 1000 gm. Find profit or loss

$$\boxed{\text{S.C. Diff} \times 100 \over \text{lower val}}$$

$$\text{Sol} \quad = \frac{100 \text{ gm}}{1150} \times 100$$

$$= 9.09\% (\uparrow)$$

$$\text{Sol} \quad \frac{100}{900} \times 100 = 11.11\% (\uparrow)$$

1. Shopkeeper labelled price of his article so as to earn profit of 30% on C.P. He then sold article by offering a 10% on labelled price. what actual % profit earned by dealer

$$\text{Sol} \quad \begin{array}{c} \checkmark 100 \\ \downarrow +30\% \\ 130 \end{array}$$

$$\text{or} = +30 - 10 - \frac{30 \times 10}{100}$$

$$= 20 - 3 = 17\%$$

$$\downarrow -10\% (130 - 10\%) \\ 117$$

$$\text{profit \%} = 17\% (100 - 117)$$

- Shopkeeper earns a profit of 25% after allowing 10% discount on list price. what is C.P. of article whose list price is 500/-

$$\text{Sol} \quad \begin{array}{c} 500/- \\ \downarrow -10\% (50) \\ 450/- \end{array} \quad \begin{array}{c} 125\% = 450/- \\ 100\% = ? \end{array}$$

- By selling 12 Note books, a merchant gains the S.P. of 4 Note books. find his gain percent

$$\text{Sol:} \quad \text{Sp of one book} = S$$

$$\text{Total Sp of 12 books} = 12S$$

$$\text{profit} = \text{Sp (4 books)} = 4S$$

$$\text{Total Cp} = \text{Total Sp} - \text{profit}$$

$$\text{Cp} = 12S - 4S = 8S$$

$$\text{profit \%} = \frac{\text{profit}}{\text{Cp}} \times 100$$

$$= \frac{4S}{8S} \times 100$$

$$\text{profit \%} = 50\%$$

Percentage

Typ 1: $x\%$ of $y = \frac{x}{100} \times y$

$y\%$ of $x = \frac{y}{100} \times x$

15% of $x = 60\%$ of 2000

$\frac{15}{100} \times x = \frac{60}{100} \times 2000$

Typ 3: $\%$ of x in $y = \frac{x}{y} \times 100$

A is $x\%$ more than to C
B is $y\%$ more than to C

$\%$ of A in B = $\frac{100+x}{100+y} \times 100$

$\%$ of B in A = $\frac{100+y}{100+x} \times 100$

A $\xrightarrow{+2\%}$
100+x

B $\xrightarrow{+y\%}$
100+y

Ex: A is 20% more than to C, B is 50% more than to C
what is $\%$ of A in B?

Sol: $\frac{120}{150} \times 100 = 80\%$

Ex: 2 No's are 15% & 20% less than a third number. what $\%$ is the first Num of second Num

Sol: ie 85, 80 $\Rightarrow \frac{85}{80} \times 100 = 106.25\%$

Typ 4: A reduction of 10% in the price of sugar enables a house holder to obtain 3kg more for 225/- the price per kg of sugar after reduction?

Sol: after reduction

10% = 3 kg

100% = 30 kg = 225

kg = $225/30 = 7.5/-$

Initial 225 = (x kg) (y)

225 = (x+3) (0.9y)

xy = (x+3) 0.9y

x = 0.9x + 2.7

0.1x = 2.7

x = 27 kg

After x+3 = 30 kg

30 kg: 225

kg = $\frac{225}{30}$

kg = 7.5/-

Typ 5: Population Concept

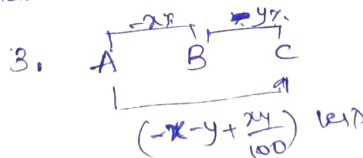
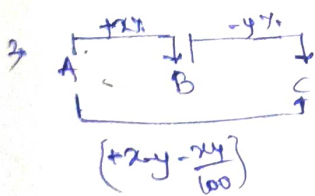
present population is given (p) which increased by $x\%$ per every annum, then population after n years is

$P = p \left[1 + \frac{x}{100} \right]^{\pm n}$
 $\downarrow$ present pop
 $\rightarrow$ After
 $\rightarrow$ ago (before)

x = rate %

n = Time (years)

Typ 8: A is $x\%$ more than B and B is $y\%$ more than to C then A is $\left[x+y + \frac{xy}{100} \right]\%$ more than C



Ex: length of rect increased by 20%, breadth decreased by 10%. Find change of its Area?

Sol: $A = l \times b$ $l = 10 \xrightarrow{+20\%} 12$

$b = 10 \xrightarrow{-10\%} 9$

$26 \times 100 = 108$

$\frac{8}{100} \times 100 = 8\%$

$+20-10 - \frac{20(10)}{100} = 8\%$

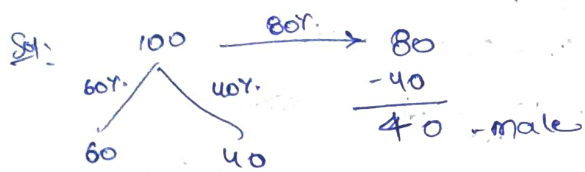
Typ 9: In an exam A & B are two students A got 30% of marks and failed by 10%, B got 40% of marks passed by 20%. Find total marks?

Sol: 10% = 30

100% = ? (300)

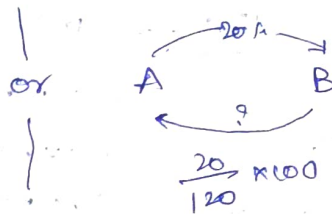
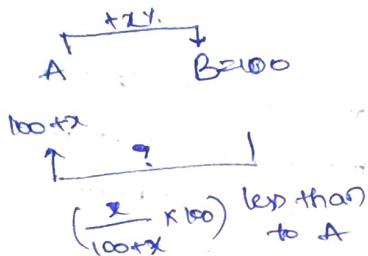
mil
 $\frac{10}{100} \times 300 = 30$
 $\frac{20}{100} \times 300 = 60$
 $10+20 = 30$

2. In party, 60% of invited guests are male and 40% are female. If 80% of Invited guests attended the party and if all the invited female guests attended, what would be the ratio of male to female among the attendees in party?

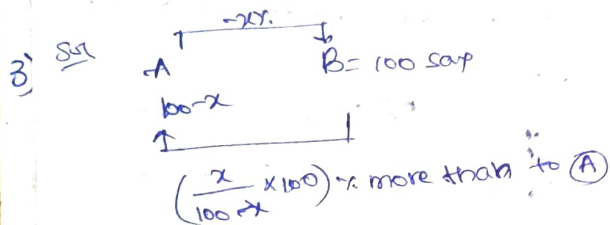


Male: female $\Rightarrow 40:40 \Rightarrow 1:1$

Typ 10: A is $x\%$ more than B then B is $\left(\frac{x}{100+x}\right) \times 100$ less than A



2. A is $x\%$ less than B then B is $\frac{x}{100-x} \times 100$ more than A



Ratio and proportion

Typ 1: Given ratio $a:b$

Duplicate ratio = $a^2:b^2$

$$= \sqrt{a}:\sqrt{b}$$

$$= 3\sqrt{a}:3\sqrt{b}$$

$$= \frac{1}{a}:\frac{1}{b}$$

Typ 3: $A:B = 2:5 \times 3$

$B:C = 3:7 \times 5$

$A:B:C \Rightarrow 6:15:35$

$A:B = 6:15$

$B:C = 15:35$

$2:5$
 $3:7$

$6:15:35$

Typ 4: proportion

$$a:b = c:d$$

$$\frac{a}{b} = \frac{c}{d}$$

$$ad = bc$$

TIPS

mixture: 50 lit of quantity mixture milk + water, milk to water

Ratio 3:2 How many lit added to given mixture then mixture 2:3

Some quantity of water added

Sol: $M:W = 3:2$

$$M = \frac{3}{5} \times 50 = 30$$

$$W = \frac{2}{5} \times 50 = 20$$

$$30:20+x = 2:3$$

$$90 = 2(20+x) \Rightarrow x = 25 \text{ lit}$$

Typ 6: 2430 has been divided among A, B, C in such a way that if their shares be reduced by 25, 20, 15 respectively. The remainders are in ratio 3:4:5 then A's share?

$$\text{Sol: } 2430 - [25 + 20 + 15] = 2400$$

3:4:5

$$A's = \frac{3}{12} \times 2400 = 600$$

$$\text{Before reduction A's share} = 600 + 25 = 625$$

If $x:y:z = \frac{1}{2}:\frac{1}{3}:\frac{1}{4}$ then, what is value of $(x+z-y)/y$? $= \frac{\frac{1}{2} + \frac{1}{4} - \frac{1}{3}}{\frac{1}{3}} = \frac{6:4:3}{12}$

A sum of money is distributed among P, Q, R, S in proportion 5:2:4:3 respectively.

If R get 1000 more than S what is share of P

$$\text{Sol } 4-3 = 1000 \Rightarrow \frac{1}{14} \times A = 1000 \Rightarrow A = 14000$$

$$\frac{2}{14} \times 14000 = 2000$$

Seats for maths, phy, chemist in a school are 5:7:8 they increased seats by 40, 50, 30% respectively, new ratio

Sol $M:P:R = 5:7:8$

$$5x:7x:8x$$

$$40\% \downarrow \quad 50\% \downarrow \quad 30\% \downarrow$$

$$700:1050:1040$$

$$70:105:104 \Rightarrow 2:3:4$$

To const wall sand & cement mixture 3:1. The cost of sand to cement is 1:2. If total cost of

sand and cement to const the wall is 1000 rupees, then cost of cement is?

Sol: $\rightarrow 3:1$

$$\text{Cement} = \frac{2}{5} \times 1000 = 400$$

$$\rightarrow \frac{1:2}{(3:2)} \rightarrow 1000$$

✓ Smallest angle of $\Delta = \frac{2}{3}$ of smallest angle in quadrilateral. The ratio b/w angles of quadrilateral $3:4:5:6$. The largest angle of $\Delta = 2 \times \text{smallest angle}$. what is sum in degree of second largest angle of Δ and largest angle of quadrilateral

Sol $3x + 4x + 5x + 6x = 360 \Rightarrow 18x = 360 \Rightarrow \boxed{x = 20}$ $(60, 80, 100, 120)$

Δ A B C
 $\frac{2}{3}(60)$ $2(40)$
 $(40) + 60 + (80) = 180$
 $\downarrow$

^{second large}
 $\Delta = 60 + 120 \rightarrow \text{largest angle quad}$
 $\Delta \text{ angle} = 180$

TYP #1 Coin Concept

1. A bag contain 1, 2, 5 $\text{\pounds}$ coins their ratio $3:2:1$ bag contains 90 coins. find no of individual coins and also find individual Amt.

Sol coins ratio $3:2:1$, 90 coins

	Ind Coin	Amt
1 $\text{\pounds}$ $\frac{3}{6} \times 90 = 45$		$45 \times 1 = 45$
2 $\text{\pounds}$ $\frac{2}{6} \times 90 = 30$		$30 \times 2 = 60$
5 $\text{\pounds}$ $\frac{1}{6} \times 90 = 15$		$15 \times 5 = 75$

1. five year ago ratio of Aman's age to his father's age was $1:4$ and five years from now the ratio will be $2:5$. what was father's age when Aman was born?

Sol: $1:4 \rightarrow x \quad 4x$ (5y ago)

$+5 \downarrow$	$\downarrow +5$
present $\rightarrow x+5$	$4x+5$
$+5 \downarrow$	$\downarrow +5$

$8x+20 = 5x+50$
 $3x = 30$
 $x = 10$

after 5 years $2+10 : 42+10 = 2:5$

	present	father
Aman	$5+10$	$4(10)+5$
	15	45
when born	Aman father age 30	

Alligation and mixture

TYPE I: Alligation rule

$$\begin{array}{r} \frac{P_1}{a/-} \\ \frac{P_2}{b/-} \\ \hline c/- \text{ [C.P of mix]} \\ b-c : c-a \end{array}$$

or

$$\begin{array}{r} a/- \quad b/- \\ \quad \diagdown \quad \diagup \\ \quad \quad c \\ \quad \diagup \quad \diagdown \\ c-b : a-c \end{array}$$

SI and CI

Diff b/w SI & CI

	SI	CI	
1 st Year	100	100	$[100 + 1000 = 1100]$
2 nd Year	100	110	$10\% = 110$
3 rd Year	100	121	$110 + 1100 = 1210$

SI is always same per every year

CI is changed per every year

SI

SI

$$SI = \frac{PTR}{100}$$

$$A = P + SI$$

P = Principal amount

T = Time (years)

R = Rate %

A = Amt

Diff Timing, R.Y. :-

$$T[SI] = \frac{P}{100} [T_1 R_1 + T_2 R_2 \dots]$$

$$A = P + T[SI]$$

CI

CI

$$A = P \left[1 + \frac{R}{100} \right]^n$$

$$C.I = A - P$$

$$A = P \left[1 + \frac{R_1}{100} \right]^{t_1} * \left[1 + \frac{R_2}{100} \right]^{t_2} \dots$$

$$C.I = A - P$$

Sc: (C.I.)

If interest cal half yearly

$$A = P \left[1 + \frac{\frac{R}{2}}{100} \right]^N \quad N \rightarrow \text{No. of half years in } \phi/s$$

If interest is cal quarterly

$$A = P \left[1 + \frac{\frac{R}{4}}{100} \right]^N \quad N \rightarrow \text{No. of quarters existing in given } \phi/s$$

3) Ex: find the CI on 16000/- at 20% annum @ 9 months compounded quarterly?

Sol

$$16000 \left[1 + \frac{20/4}{100} \right]^3 = 18522$$

$$C.I. = 18522 - 16000 = 2522/-$$

Summary

Trains

Time to cross the object = $\frac{L(T) + L(obj)}{S(T)}$

Ratio
Time = $\frac{L(T_1) + L(T_2)}{S(T_1) + S(T_2)}$ $\xrightarrow{T_1} \xleftarrow{T_2}$

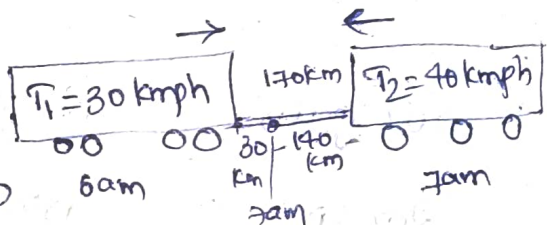
present
Time = $\frac{L(T_1) + L(T_2)}{S(T_1) - S(T_2)}$ $\xrightarrow{T_1} \xleftarrow{T_2}$

Police catching thief
Time = $\frac{\text{Total dist}}{\text{Relative Speed}}$

= $\frac{60 + 20}{40 - 20}$

Trains meet
Time = $\frac{\text{dist b/w train}}{\text{Sum of speeds}} = \frac{140}{30 + 40}$

= 2hr
So Trains meet at 7am + 2hr = 9am



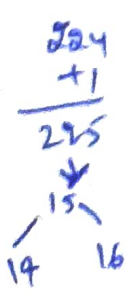
Boats & streams

Down stream speed (DSS) = $x + y$ km/h

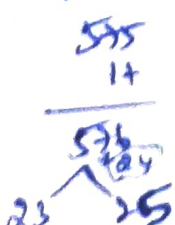
Up stream speed (USS) = $x - y$ km/h

Total dist travelled. A $\xrightarrow{x+y, D, t_1}$ B $\xleftarrow{x-y, D, t_2}$
 $T = t_1 + t_2$
 $= \frac{D}{x+y} + \frac{D}{x-y}$

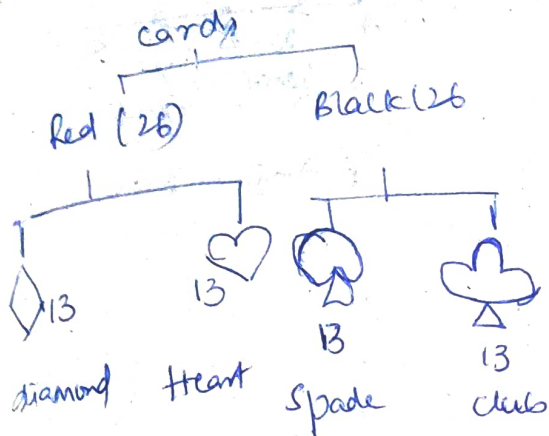
prod of two consecutive two even numbers is 224 find the Num?



prod of two consecutive odd Num 575



Probability



3 card prob

$$2 \text{ clubs \& 1 \& card} = \frac{{}^{13}C_2 \times {}^{13}C_1}{52C_3}$$

Dice

Total No. of changes = 6

$n = \text{No. of die}$

Coin

$$\text{Probability} = \frac{nCr}{2^n}$$

Basic formula

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$A \cup B = A + B - A \cap B$$

A & B are Independent events $P(A \cap B) = P(A) \cdot P(B)$

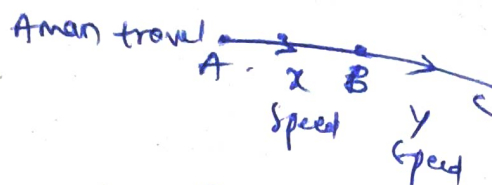
A & B are mutually exclusive events then $P(A \cap B) = 0$

Speed & distance

$$\text{Avg speed} = \frac{\text{Total dist}}{\text{Total time}}$$

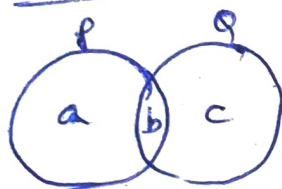
$$\text{If two dist same} = \frac{2xy}{x+y}$$

Avg speed



$$\text{If } A \rightarrow B = B \rightarrow C$$

Venn diagrams



$$P = a, b$$

$$Q = b, c$$

$$P \cap Q = b$$

$$\text{only } P = a$$

$$\text{only } Q = c$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$$

Time and Work

man days hours

$$M_1 D_1 H_1 x = M_2 D_2 H_2 y$$

↓
n

$$\text{A can do work in 10 days} \rightarrow A's \text{ 1 day work} = \frac{1}{10}$$

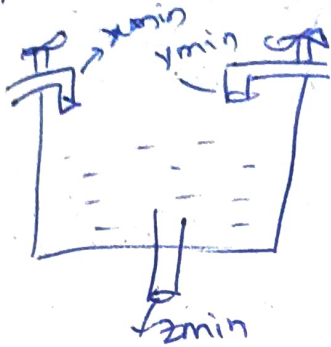
$$A \rightarrow 10 \text{ days}$$

$$B \rightarrow 20 \text{ day}$$

$$A \& B \rightarrow ? \quad \frac{1}{10} + \frac{1}{20} = \frac{1}{10}$$

①
2 in together
day work

Pipes & Cisterns

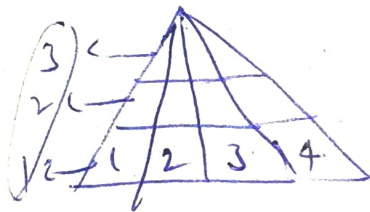


$$\frac{1}{x} + \frac{1}{y} - \frac{1}{z} = \frac{1}{\text{no. of min}}$$

No. of Square

→ No. of Horizontal lines

$m_1 \times n_1 \rightarrow$ No. of vertical lines



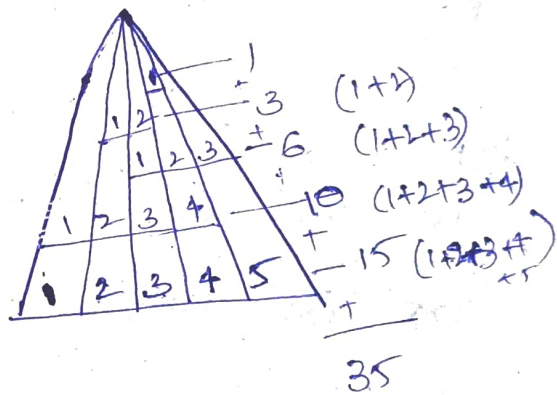
$$(1+2+3) \times 3$$

+ 7

Joining mid point



only staying one



profit & losses

$$\text{profit} = \frac{P}{C_p} \times 100$$

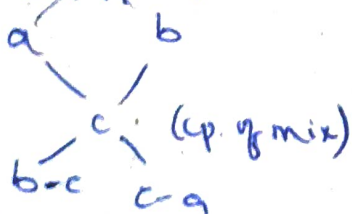
$$\text{loss} = \frac{L}{C_p} \times 100$$

Challenger

$$Q = \left(\frac{1}{2} m - 30 \right)$$

Allegation & mixtures

Type 1 Type 2



SI

$$SI = \frac{PTR}{100}$$

$$A = P + SI$$

C.I.
rupee $\rightarrow$ rate (%)
time (years)

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$$C.I = A - P$$

$$AP = a, a+d, a+2d$$

$$GP = a, ar, ar^2$$

Harmonic progression $= \frac{1}{AP}$

HCF

15 75 60

1×5 3×5 $25 \times 3 \times 5$

$$HCF = 3 \times 5 = 15$$



$$\text{Area} = \frac{\sqrt{3}}{4} a^2$$

Time and distance

$$\text{Speed}(S) = \frac{\text{Distance}(D)}{\text{Time}(T)}$$



Typ-I: A man running 10km/h, Took 5min rest for every 1km Total time

for 5km



$$10\text{km} = 60\text{min}$$

$$5\text{km} = ? \quad 30\text{min} \Rightarrow 30\text{min} + 20 \Rightarrow 50\text{min}$$

Typ-II: Avg

Blood relations

भाई बहन - Nephew

~~बहन भाई~~ - Niece
दाughter

male +
female -

couple $\square$

Parent
child

p
|
child

Bro, sis. " = "

B = S, S = B

Father side - paternal

Mother side - maternal

Num sys
 1) Divisible by 3 $\rightarrow$ sum of numbers divisible by 3 the total num
 by 9 $\rightarrow$ its divisible by 3

divisible by 6 : It must be divisible by 2 & 3

by 4: Check last two num only if divisible then num divisible by 4.

1236 $\rightarrow \frac{36}{4} = 9$ so 1236 is divisible by 4

LCM

$$\begin{array}{r|l} 3 & 12, 15, 18 \\ \hline 2 & 4, 5, 6 \\ \hline & 2 \quad 5 \quad 3 \end{array}$$

(or)

$$\begin{array}{ccc} 12 & 15 & 18 \\ \wedge & \wedge & \wedge \\ 3 \times 2^2 & 3 \times 5 & 3^2 \times 2 \end{array}$$

$$3^2 \times 2^2 \times 5 = 180 \text{ LCM}$$

$$3 \times 2 \times 2 \times 5 \times 3$$

$$= 180$$

HCF

$$\begin{array}{ccc} 15 & 45 & 60 \\ \wedge & \wedge & \\ 3 \times 5 & 3^2 \times 5 & 2^2 \times 3 \times 5 \end{array} \quad \text{least power}$$

$$3^1 \times 5^1 = 15 \text{ HCF}$$

• product of 2 numbers = product of HCF and LCM

$$x \times y = \underset{(HCF)}{LCM} \times \underset{(LCM)}{HCF}$$

• No. of prime factors

$$\begin{array}{cccc} 6^{10} & 7^{14} & 11^{27} & \\ \textcircled{2} \textcircled{3} & \textcircled{7} & \textcircled{11} & \textcircled{27} \\ 2 & 3 & 7 & 11 \end{array}$$

$$\Rightarrow 10 + 10 + 17 + 27 = \textcircled{64} \text{ no. of prime factors}$$

sol:

Cyclicity

Cyclicity of 2 = 4

$$2^{103}$$

Sy $\frac{103}{4}$ we need remainder

$$2^3 = 8 \text{ so}$$

$$2^{103} = 8 \text{ unit digit (or) last digit}$$

$$2^1 = 2$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

$$8$$

$$4) 103 = 25$$

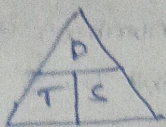
$$\frac{8}{23}$$

$$\frac{20}{20}$$

$$\textcircled{3} \text{ remainder}$$

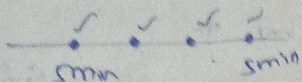
Time and distance

$$\text{Speed}(S) = \frac{\text{Distance}(D)}{\text{Time}(T)}$$



Q-1: A man running 10km/h, took 5min rest for every 4km Total time

At 5km



$$10 \text{ km} = 60 \text{ min}$$

$$5 \text{ km} = ? \quad 30 \text{ min} \Rightarrow 30 \text{ min} + 20 \Rightarrow 50 \text{ min}$$

Q-2: Avg

Blood relations

ಪುರುಷರು - Nephew

ಮಹಿಳೆ - Niece

ಮಕ್ಕಳು
daughters

male +

female -

Couple A

Parent
child

P
|
Child

Bro, sis. " "

B = S, S = B " "

Father side - paternal

Mother side - maternal

Cyclicity

cyclicity of 2 = 4

$$2^1 = 2$$

$$2^5 = 32$$

$$2^{103}$$

$$2^2 = 4$$

$$2^3 = 8$$

$$2^4 = 16$$

So $\frac{103}{4}$ we need remainder

$$5 =$$

$$2^3 = 8 \text{ so}$$

$$2^{103} = 8 \text{ unit digit (or) last digit}$$

$$4 \overline{) 103} (25$$

$$8$$

$$\underline{23}$$

$$20$$

$$\underline{3}$$

remainder